

Standard Operating Procedure

|                  |                                                        |           |   |
|------------------|--------------------------------------------------------|-----------|---|
| Document Number: | SOP-BR-014                                             | Revision: | 2 |
| Title:           | Time-Course Bioreactor Sampling and Viability Analysis |           |   |
| Effective Date:  | May 1, 2026                                            |           |   |

Trellis Bioworks    Project: CHO-K1 Fed-Batch Optimization    Generated: May 17, 2026

1.0 Purpose

Standardise the time-course sampling and in-process viability analysis of CHO-K1 fed-batch cultures across the 2 L Sartorius Biostat platform so that inter-operator and inter-batch results are directly comparable.

2.0 Scope

Applies to all GLP fed-batch development runs performed in Suite 2B on the Biostat A 2 L vessels (BR-201 through BR-204). Does not cover seed-train shake-flask sampling.

3.0 Procedure: Time-Course Bioreactor Sampling and Viability Analysis

**CRITICAL REQUIREMENT:** All sampling MUST occur within  $\pm 5$  min of the nominal time point. Deviations  $> 5$  min require an in-process deviation (IPD) report before the next time point.

3.1 Hour 0 (T=0) — Inoculation Sampling

Reference time point. Sample drawn immediately after inoculation establishes baseline viability and metabolite concentration.

Figure 1: Inoculation sampling port (post-CIP).

Annotated photograph of the bioreactor sampling port showing the sterile septum and integrated luer fitting.

| Time Target | Action                                                                           | Expected Output / Log                          |
|-------------|----------------------------------------------------------------------------------|------------------------------------------------|
| T=0 min     | Aseptically draw 5 mL via sterile sampling port into pre-labelled 15 mL conical. | Sample S-0 logged in eBR; cap torque verified. |
| T+10 min    | Perform trypan-blue viability count                                              | Viable cell density (cells/mL);                |

|                 |                                                         |                                                             |
|-----------------|---------------------------------------------------------|-------------------------------------------------------------|
|                 | (Cellometer K2, 1:1 dilution).                          | viability %.                                                |
| <b>T+15 min</b> | Submit retained aliquot to in-process QC (pH, glucose). | pH, glucose (g/L) recorded; sample chain-of-custody signed. |

### 3.2 Hour 3 (T=3) — Foam Growth Check

Inspection only — no draw unless foam height exceeds the documented limit.

**Figure 2: Acceptable vs. excessive foam.**

*Side-by-side comparison of acceptable headspace foam (10 mm) and the rejection threshold.*

| Time Target         | Action                                                                             | Expected Output / Log                                  |
|---------------------|------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>T=3 h</b>        | Visually inspect headspace foam height; compare to Figure 2.                       | PASS / FAIL noted in eBR; foam height (mm).            |
| <b>T=3 h +5 min</b> | If FAIL: add 0.5 mL sterile antifoam-C via dedicated addition port and re-inspect. | Antifoam lot, volume, and post-add foam height logged. |

### 3.3 Hour 6 (T=6) — Feed Initiation Sampling

Sample drawn immediately prior to bolus feed addition; establishes the pre-feed metabolic profile.

| Time Target          | Action                                                                                       | Expected Output / Log                             |
|----------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>T=6 h</b>         | Aseptically draw 10 mL into pre-labelled tube; split into viability and metabolite aliquots. | Sample S-6 logged; aliquots routed to bench + QC. |
| <b>T=6 h +20 min</b> | Initiate bolus feed (Feed-A, 50 mL) via peristaltic pump P-02.                               | Feed start time, pump RPM, and totaliser reading. |